Workflow for Oil Well Performance Analysis (a case study review)

Jeremiah Ifeanyi OKOROMA

Paris, France

B.Tech. – Petroleum Engineering

MSc. – Reservoir Engineering

jeremiah.okoroma@hotmail.com

About the Author

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Case Study

An oil reservoir, discovered in the North Sea. A vertical well has been drilled with single point fluid entry and perforation type completions at reservoir pressure and temperature of 3600 psi and 200 °F respectively. There are no isochronal or multipoint well test data; however, a 4-hour well/production test was run and a rate of 9000 STB/d was obtained at a flowing wellhead pressure of 300 psi. Laboratory fluid analysis has been performed and stock tank properties and other data relevant for model calibration have been obtained. Reservoir simulations have been performed to predict the change in water cut over the field life (to be consolidated by water cut sensitivity analysis). Although the preferred reservoir pressure maintenance is water injection, a feasibility study on the use of artificial lift methods is required to assess the benefits of using gas lift and ESP. It has also been revealed from log analysis that a shallow gas zone exists up-hole from the producing zone. Consequently, a self-lifting feasibility study is proposed.

Well configuration

Tubulars	Casing	Tubing
Bottom MD (ft)	9000	8600
Internal diameter (in)	7.511	3.958
Wall thickness (in)	0.557	0.271
Roughness (in)	0.001	
Deviation survey data		
MD (ft)	**TVD** (ft)	
0, 1000, 2500, 5000, 7500, 9000	0, 1000, 2450, 4850, 7200, 8550	
Completions & downhole equipment		
Completion geometry	Vertical	
Fluid entry type	Single point	
Perforation MD (ft)	8800	
Packer's MD (ft)	8500	

Heat transfer data	
U-value (Btu/h/ft²/°F)	5
Ambient temperature @ measured depths	50 °F @ 0 ft 200 °F @ 9000 ft

Laboratory fluid analysis data

Bubble point properties	
Pressure (psia)	2100
Temperature (°F)	200
Solution gas (scf/stb)	500

Stock tank oil properties	
GOR (scf/stb)	500
Water cut (%)	10
Gas specific gravity	0.8
Water specific gravity	1.05
Oil density (API)	36

Black Oil calibration data	
Oil FVF (above P_b)	1.18 @ 3000 psi and 200 °F
Oil FVF (below P_b)	1.22 @ 2100 psi and 200 °F
Dead oil viscosities	1.5 cP @ 200 °F, 10 cP @ 60 °F
Live oil viscosity	1.1 cP @ 2100 psi and 200 °F
Gas viscosity	0.029 cP @ 2100 psi and 200 °F
Gas compressibility	0.8 @ 2100 psi and 200 °F

Supplementary data

Field & asset constraints (where applicable)	
Gas lift availability (MMSCFD)	*3*
Design liq. Prod. rate (STB/d)	*10000*
WCUT max (%) – Res. simulation	*70%*
Surface gas prod./well (MMSCFD)	*6*

Data for possible gas lift design	
Gas lift installation MD (ft)	*8000*
Gas specific gravity	*0.64*
Surface injection tmperature (°F)	*100*

Data for possible self-lifting	
Reservoir pressure (psia)	*3200*
Reservoir temperature (°F)	*180*
Completion MD (ft)	*8000*

Step 1: Development of the best case model

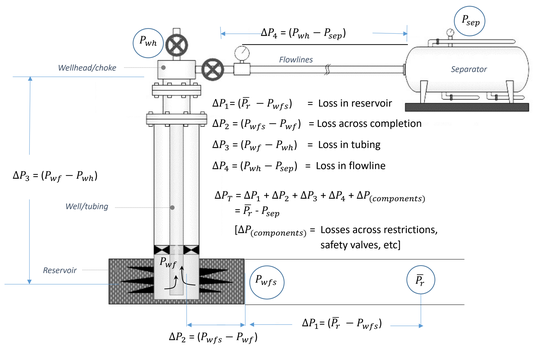
1.1 Initial well design and Nodal Analysis

Nodal analysis is used to evaluate the performance of a production or injection well, particularly the fluid dynamics within the near wellbore zone (reservoir) and the tubulars (vertical and horizontal).

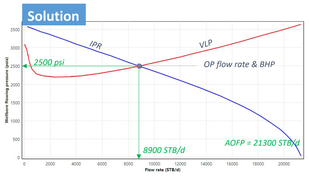
Basically, It involves identifying a nodal point (also known as solution node), that splits the producing system into two parts: the inflow (IPR) and the outflow (VLP). This solution node could be the wellhead or bottomhole.

The purposes of Nodal Analysis include, but are not limited to:

1. To determine the operating conditions (flow rate and BHP) of a well taking into consideration the wellbore geometry and completion constraints; as well as the likely conditions the well may die or experience liquid loading.
2. Based on the aforementioned operating conditions, optimize the production system for the best profitable scenario (case).
3. To investigate and select an optimal lift method, as well as adequate period for installing such artificial lifts.
4. To perform dedicated item checks and identify unwanted flow restriction culprits.
5. To perform brownfield/greenfield design of production system components: tubing/flowline sizing, choke sizing, well stimulation assessment, etc.



Solution



Nodal Analysis performed with bottomhole as solution node, and outlet pressure as 300 psi.

Consider that the nodal point is identified as bottomhole, the principles of Nodal Analysis include, but are not limited to:

1. Well operating point is the intersection of the IPR and VLP curves, it represents the point at which the law of continuity is fulfilled and provides the flow rate and BHP.
2. Highest obtainable production flow rate is possible when the pressure of solution node (BHP) is equal to zero. This is the theoretical maximum a reservoir/completion can deliver, known as the Absolute Open hole Flow Potential (AOFP).

1.2 PVT fluid calibration with laboratory data

The operating point flow rate predicted by Nodal Analysis (and dedicated software – Prosper, PIPESIM, etc.) is only an estimate containing intrinsic uncertainty, based on constraints associated to data quality and the models (correlations for multi phase flow) used. It goes without saying that the better the data and model quality, the better the predicted rates and flowing pressures. Hence, in order to obtain the best case model, increase the accuracy of fluid properties calculations, and improve overall model predictability, it is important to tune PVT correlations to match laboratory measured data (if available); as well as to calibrate multiphase flow correlations or other models with field measurements.

The adverse effects of no or poor PVT fluid calibration include:

- Calibration of the bubble point pressure can result in the initial appearance of gas at a depth of 100's of meters away from that obtained from an uncalibrated model. This results in a significantly different fluid mixture density and, thus, a vastly different elevational pressure gradient.
- Calibration of the fluid viscosity can drastically improve the calculation of the frictional pressure gradient, particularly in heavy oils and emulsions.

Solution

	Before calibration	After calibration
Production Rate (STB/d)	8900	8211.48
Flowing BHP (psi)	2500	2573.56
Flowing WHT (°F)	141	140.25
AOFP (STB/d)	21300	21319.90
Gas appearance depth (ft)	7091 – 7213	6848 - 6965

For an enhanced model predictability, the overall heat transfer coefficient may also be tuned to match field measurements (BHP, rates and temperatures).

1.3 IPR matching (well test data or PI sensitivity analysis)

In addition to PVT calibration, the modelling and simulations results can be further improved by calibrating the well IPR equations with flowing well test data from field measurements (if available). There exist more than one approach:

- Calibration based on multipoint or isochronal well test data.
- Calibration based on well performance data (rate, static reservoir pressure, and flowing wellhead pressure).

For the present case study, the IPR is matched via sensitivity analysis on PI, that is, iteratively calculating the PI value (exploiting – for instance – the direct proportionality relationship between PI and rate) that matches the known flow rate and boundary conditions.

Solution

	Before IPR matching	After IPR matching
Productivity index (STB/d/psi)	8	9.38
AOFP (STB/d)	21320	24850

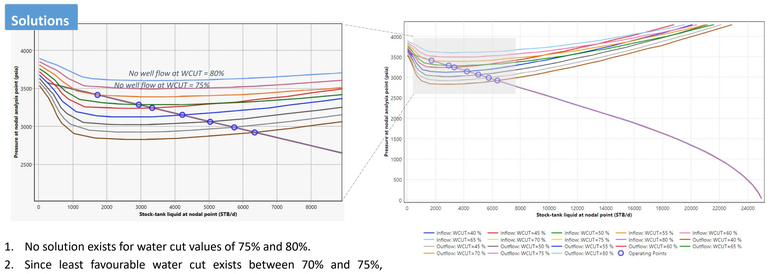
PI (in STB/d/psi) – a ratio of total liquid surface flowrate to pressure drawdown – accounts for the well's potential to produce.

Step 2: Sensitivity analysis on water cut

Increasing water production from an oil field is an inevitable factor to consider in well design. Production profile/plan obtained from reservoir simulation could help provide the annual percent of water cut associated with the oil production, throughout the field life. In order for oil production to be economically feasible and technically possible (considering natural lift), the critical water cut value must be investigated.

Beyond the critical water cut, the produced water loads (under gravity) and forms a column within the well, hydrostatic pressure losses increase and the reservoir pressure becomes insufficient to lift all liquid to the surface. For this case study and task, Nodal Analysis is used.

Solutions



1. No solution exists for water cut values of 75% and 80%.
2. Since least favourable water cut exists between 70% and 75%, further sensitivity analysis (with steps of 1%) was performed, results shows critical water cut = 71%, above which well stops flowing.

Step 3: Comparative analysis on well performance (artificial lift)

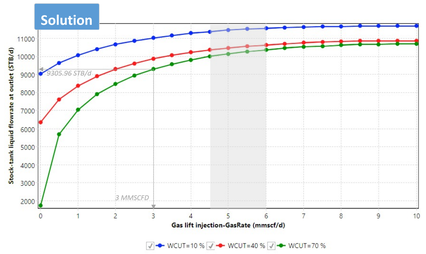
As earlier stated, reservoir pressure management for the present case study envisages water injection to support oil production rates in later field life. However, this section reviews the feasibility of using artificial lift methods: gas lift and electric submersible pumps (ESPs) methods. Then, the performances of both methods are evaluated and compared on the basis of their performance around an already defined constraint (gas lift availability = 3 MMSCFD, design liquid production rate = 10,000 STB/d @70% WCUT).

Artificial lift system are generally aimed at maintaining a reduced bottom hole pressure so as to allow the oil (or whatever desired reservoir fluids) to be produced at an economically feasible rate. The main mode of operation for most artificial methods is either of the following: reducing hydrostatic head or lifting the fluids via mechanical device. While gas lift applies to the former; ESP, rod pumps and progressive cavity pumps (PCPs) provides the later.

3.1 Gas lift injection

Basically, gas lift injection reduces the tubing fluid density and hydrostatic component of pressure gradient above the point of gas injection, thereby reducing the back pressure at the completion. The reduced bottomhole pressure effectively enables more reservoir fluids to flow

Solution



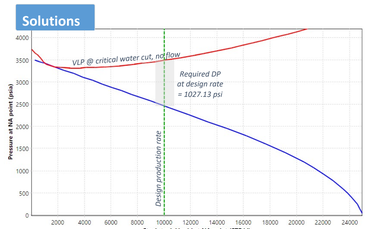
1. Given the following gas lift system constraint (gas lift availability = 3 MMSCFD, design liquid production rate = 10,000 STB/d @70% WCUT).
2. Available gas injection rate is not able to produce as expected since only 9305.96 STB/d is possible.
3. Gas injection rates between the range of 4.5 to 6 MMSCFD are suitable; below that, production is less than required.
4. Further incremental change above suitable range of injection rate may yield incremental liquid production rates that are non-economical (as curve gets less steep).
5. Gas lift system is **not able to deliver target production rate**.

3.2 ESP design, installation and performance

An electrical submersible pump (ESP) is a multistage centrifugal pump that is able to provide a substantial increase in pressure to produce large volumes of reservoir fluids. The main components found in one ESP stage are impeller and a diffuser mounted on a shaft. The working mechanism involves imposition of kinetic energy on the fluid due to centrifugal forces created by impellers attached to the rotating shaft, diffusers then convert the fluid's kinetic energy to pressure, enabling the fluid to travel to the surface.

The design of an ESP take into account the casing internal diameter, equipment clearance, and design production rate at pump intake conditions.

Solutions

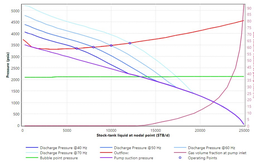


Initial Nodal Analysis showing artificial lift design conditions

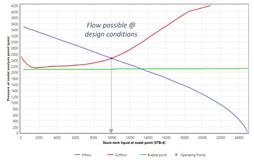
1. Given the artificial lift design constraint: liquid production rate = 10,000 STB/d @70% WCUT).
2. The following highlights the ESP design result:

No. of stages	68
Efficiency (%)	72.74
Power (hp)	255.45
Speed (rpm)	3499.99
Head (ft)	2487.09
Differential pressure (psi)	1018.02
Fluid temperature rise (°F)	1.41

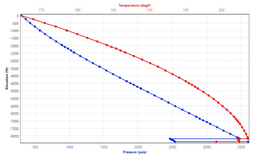
3. Gas volume fraction (GVF) is zero at pump intake conditions because the fluid is above the bubble point (no free gas).
4. ESP is **able to deliver target production rate**.



ESP well performance



Final Nodal Analysis based on ESP design



Well P-T profile based on ESP design

3.3 Self-lifting the well (multiple completions)

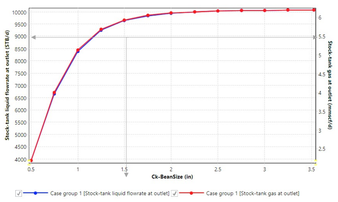
By perforating a shallow gas zone located up-hole of the already producing zone, it may be possible to self-lift the well, as an alternative to gas lift and ESP systems. Downhole equipment necessary for the new completion include a single packer to divert gas flow from new perforation into the tubing, through a sliding sleeve. In addition, the reservoir performance is modelled using the pseudo-steady state Darcy IPR equation based on the reservoir properties provided in table below.

Reservoir pressure (psia)	3200
Reservoir temperature (°F)	180
Completion MD (ft)	8000
Reservoir thickness (ft)	5
Borehole diameter (in)	12.25
Reservoir permeability (mD)	20
Mechanical skin	0
Rate dependent skin	0

Solution

(oil zone @ 70% water cut)	Before Self-lifting	After Self-lifting
Liquid production late (STB/d)	1750	10076.26
Stock-tank gas rate (MMSCFD)	1.5114	6.2016
Gas contribution from upper gas zone (MMSCFD)	4.6902	

Self-lifting the well via multiple completions is **able to deliver target production rate**.



Considering that the surface production facility has a gas production constraint of 6 MMSCFD per well, compared with the total stock-tank gas rate of 6.2016 MMSCFD due to gas contribution from upper gas zone. It becomes imperative to control the gas flow from at least the upper completion, this is accomplished using a downhole choke. For that, a sensitivity analysis is performed for a range of choke bean size (orifice sizes) from 0.5 to 3.525 inches.

(oil zone @ 70% water cut)	Before choking	After choking
Liquid production late (STB/d)	10076.26	9643.91
Stock-tank gas rate (MMSCFD)	6.2016	5.9315
Gas difference (MMSCFD)	0.2701	
Suitable bean size (in)	1.5	

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